

1. (4 pts) The Doppler-broadened linewidth of the He-Ne laser is 1.5 GHz. What is the corresponding linewidth in wavelength? (632.8 nm for He-Ne)

2. (4 pts) Sketch the spectrum you measured for the room fluorescent lights. Based on your measured spectrum for the Hg/Ar calibration source, what are the narrow spectral peaks in the room light spectrum due to?

3. (4 pts) How did the monochromator slit width affect your measured spectrum for the HeNe laser? Sketch the spectrum you measured using 50um slits and 10um slits.

4. (4 pts) Sketch the spectra you recorded for the white and blue LED's. Label on your spectra the "linewidths" and label the center wavelength and linewidth values you measured.

5. (4 pts) Do you think that the linewidth for the HeNe laser that you measured with the finest smallest slit possible represents the "true" linewidth of the laser? Explain.